

System for High Integrity Elimination of Low-Altitude Drones (SHIELD)

Requirements Document

1. Project Title, Team Members

System for High Integrity Elimination of Low-Altitude Drones (SHIELD)

- Bryson Noble (SWE) - bnoble2023@my.fit.edu
- Sophia Versage (AEE) - sversage2024@my.fit.edu
- Edward Ledwith (AEE) - eledwith2023@my.fit.edu
- Nick Jacobson (AEE) - njacobson2023@my.fit.edu
- Aryaa Vijay (AEE) - avijay2024@my.fit.edu
- Miguel Mohammed-Mueses (AEE) - mmohammedmue2023@my.fit.edu
- Isabelle Chamberlain (AEE) - ichamberlain2023@my.fit.edu
- Joshua Huser (AEE) - jhuser2023@my.fit.edu
- Will Emerson (AEE) - wemerson2023@my.fit.edu
- Tharushi Pelendagama (AEE) - tpelendagama2023@my.fit.edu

2. AEE Faculty Advisor

Madhur Tiwari, PhD - mtiwari@fit.edu

3. Client

Firat Irmak, PhD - Aerospace Engineering Department, Florida Institute of Technology

4. Required Features

- **Target detection:** find targets in each camera frame within the operational range using a custom model.
- **Target tracking:** label and track targets separately using ByteTrack with a Kalman filter to ensure that each potential target is recognized as being unique.
- **Confidence monitoring:** monitor object detection confidence continuously before and during target engagement.
- **Real-time processing:** target detection and image processing using onboard computation maintaining at least 30 frames per second.
- **Ground Control System (GCS) command and control:** accept Engage and Abort commands from the GCS and report health through a heartbeat.

5. Functional Requirements

Number	Requirement	Parent Reqt	Requirements Rationale	Verification Method	Verification Strategy, other notes
AVS.01	The drone shall be capable of detecting a target within 20 meters [1-2]	SYS.03	FPV drones carry 0.3–5 kg of explosives, and the DHS evacuation distance for a 5 lb explosive is ~21 m.	<i>Analysis, Test</i>	Run automated test script and review logs to verify requirement on SW and HW.
AVS.02	The drone shall recognize and track a target with a minimum span of 0.25 m [3-7]	SYS.03	The most commonly used kamikaze drones have a span of ~0.25 m or larger.	<i>Analysis, Test</i>	Run automated test script and review logs to verify requirement on SW and HW.

AVS.01 – The Department of Homeland Security recommends a mandatory evacuation distance of ~21.34 meters (published as 70 feet) for a threat with an explosive capacity of ~2.27 kilograms (published as five pounds), and modern FPV drones typically carry between 0.3 kilograms and 5 kilograms of explosives [1][2]. Therefore, SHIELD must be capable of detecting a target within a distance of 20 meters.

AVS.02 – Most modern kamikaze drones are sold in platform sizes ranging from ~0.25 m (10-inch platform) to ~0.46 m (18-inch platform) [3-7]. Therefore, SHIELD must be capable of detecting targets with a span of 0.25 m (10 inches) or larger.

6. Performance Requirements

Number	Requirement	Parent Reqt	Requirements Rationale	Verification Method	Verification Strategy, other notes
AVS.03	The drone shall maintain at least 25 percent confidence while en route to the target [8].	SYS.03	Provides continuous verification after the operator approves engagement, allowing an abort on a false positive if the track degrades.	<i>Demonstration, Test</i>	Run an automated test script and conduct a test flight to verify requirement.
AVS.04	The drone's onboard computing system shall process target detection within 33.33 ms of receiving the data from the onboard vision system [9]	SYS.03	Real-time detection is particularly important for an effective C-UAS. This will ensure that the system is acting on current information.	<i>Analysis, Test</i>	Run automated test script and review logs to verify requirement on SW and HW.

AVS.03 – Object detection models are most commonly deployed with a confidence threshold of 0.25 to balance precision, the share of detections that are real targets (high precision means few false positives), with recall, the share of real targets that are detected (high recall means few false negatives) [8]. This number ensures that SHIELD will detect a sufficient proportion of true drone threats without generating an excessive rate of false alarms.

AVS.04 – Real-time detection is particularly important for an effective C-UAS, which requires a minimum processing rate of 30 frames per second [9]. There are 1000 milliseconds in one second, and dividing 1000 by 30 yields a maximum frame processing time of ~33.33 milliseconds to enable real-time detection.

7. Interface Requirements

Number	Requirement	Parent Reqt	Requirements Rationale	Verification Method	Verification Strategy, other notes
AVS.05	The drone shall receive engage and abort commands from the Ground Control System [10].	SYS.01	A "Control station" is defined in 14 CFR Part 107, and there must be a remote pilot in command.	<i>Demonstration</i> <i>, Test</i>	Run an automated test script and conduct a test flight to verify the requirement.
AVS.05.01	The drone shall send a heartbeat to the Ground Control System at a rate of 1 Hz [11-13]	AVS.05	The Ground Control System must have the status of the drone at all times. 1 Hz is the standard rate.	<i>Analysis,</i> <i>Test</i>	Run automated test script and review logs to verify requirement on SW and HW.

AVS.05 – The Federal Aviation Administration mandates that a remote pilot be in command of an autonomous drone and defines a control station in 14 CFR Part 107 [10]. The SHIELD Ground Control System will enable a pilot-in-command to send engagement and abort commands to the SHIELD drone to comply with federal regulations.

AVS.05.01 – To ensure that a communication loss between the Ground Control System and drone is identified promptly, the drone shall send a heartbeat to the Ground Control System at a rate of 1 Hz, which is the standard rate for telemetry links [11-13].

8. Bibliography

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